

Luca Zanolini

Research Scientist & Engineer · Distributed Systems & Reliability

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lucazanolini.com | [DBLP](#) | [Google Scholar](#) | [GitHub](#)

PROFILE

I am a distributed-systems researcher and engineer: I design protocols, prove them correct, and follow them into specifications, implementations, and tests. At the Ethereum Foundation my work spanned this full range – co-designing consensus protocols and confirmation rules that shape Ethereum’s roadmap, and leading a security analysis of EIP-7732 that uncovered specification bugs, since fixed upstream. I build systems in Rust – distributed protocols from replicated logs to Byzantine broadcast, released as an open-source course – and am now applying the same reliability discipline to multi-agent AI infrastructure. Trained as a mathematician; PhD in Computer Science from the University of Bern; 20+ peer-reviewed publications, including at ACM CCS, PODC, and IEEE ICDCS.

EXPERIENCE

Research Scientist

2022-2026

Ethereum Foundation, Protocol Consensus

Seoul / Remote

- Led the formal security analysis of EIP-7732 (enshrined proposer-builder separation) end to end: scoped the program, built open models and proofs against the executable specification, and delivered observable security guarantees under explicit adversarial and network assumptions; the analysis surfaced concrete specification bugs, since addressed and fixed through `consensus-specs` pull requests.
- Co-designed consensus protocols and confirmation rules (3-Slot Finality, fast confirmation, accountable liveness) and carried them beyond the papers: co-authored the fast confirmation rule and contributed to its specification and testing, and analyzed interactions among concurrent upgrades (ePBS, FOCIL, PeerDAS).
- Prioritized research directions with protocol teams by making trade-offs explicit, relating safety, liveness, latency, accountability, and dynamic participation to design and roadmap decisions.
- Coordinated the EF Academic Grants Round as its main technical contact, working with institutions including Duke University, Stanford, a16z crypto, and IC3 across time zones; co-organized research workshops; mentored interns and graduate researchers; communicated results through research notes, blog posts, and talks worldwide.

Visiting Scholar

2025-2026

University of Cambridge, Department of Computer Science and Technology

Cambridge, UK

- Weekly small-group supervision for Concurrent and Distributed Systems; invited seminar and guest teaching on blockchain and Ethereum.

Research and Teaching Assistant

2019-2023

University of Bern, Cryptology and Data Security Group

Bern, Switzerland

- Developed theory and algorithms for asymmetric trust, Byzantine quorum systems, and consensus; organized PhD-group seminars and research collaborations; ran tutorials, exercises, and assessment for Cryptography and for Privacy and Data Security.

Research Intern

Summer 2021

Chainlink

Remote

- Built Go research prototypes and participated in distributed-systems and blockchain research seminars.

Ethereum Developer

2017-2019

Interlogica

Italy

- Designed and implemented Solidity smart contracts in production; earned the ConsenSys Ethereum Developer Certification.

EDUCATION

PhD in Computer Science, cum laude

2019-2023

University of Bern – Advisor: Prof. Christian Cachin

Bern, Switzerland

MSc in Mathematics, 110/110 cum laude

2015-2017

Catholic University of the Sacred Heart

Italy

BSc in Mathematics, 110/110

2012-2015

Catholic University of the Sacred Heart

Italy

RESEARCH IMPACT, TECHNOLOGY TRANSFER, AND OPEN SOURCE

- [Formal security analysis of EIP-7732/ePBS](#): open models, proofs, and specification analysis for a major planned Ethereum upgrade; findings led to upstream specification fixes.
- [Concurrent and Distributed Systems in Rust](#) (2026-): an independent open-source course-in-a-repository developing distributed-systems theory through working Rust implementations, from replicated key-value stores and Raft to Byzantine broadcast, with scripted failure demonstrations and exercises.
- [Reliable Distributed Agent Systems](#) (2026-): an open-source Rust project applying distributed-systems rigor to multi-agent AI infrastructure: a replicated coordination plane with sandboxed agent workers treated as untrusted clients.
- 3-Slot Finality and fast confirmation rules: co-authored protocol designs and analyses shaping Ethereum’s consensus roadmap toward faster finality.
- Public research communication: [Finality Is in the Eye of the Beholder](#) (2025, with Vitalik Buterin and Ittai Abraham); [Why Ethereum Needs a Dynamically Available Protocol](#) (2025); Ethereum research notes on 3-Slot Finality, fast confirmation, and upgrade interactions (2023–2025).

SELECTED PUBLICATIONS

Seven of 20+ peer-reviewed papers; full list on [DBLP](#) and [Google Scholar](#).

Accountable Liveness

Andrew Lewis-Pye, Joachim Neu, Tim Roughgarden, **Luca Zanolini**. ACM CCS 2025. [doi](#)

DAG-based Consensus with Asymmetric Trust

Ignacio Amores-Sesar, Christian Cachin, Juan Villacis, **Luca Zanolini**. ACM PODC 2025. [doi](#)

TOB-SVD: Total-Order Broadcast with Single-Vote Decisions in the Sleepy Model

Francesco D’Amato, Roberto Saltini, Thanh-Hai Tran, **Luca Zanolini**. IEEE ICDCS 2025. [doi](#)

Data Independent Order Policy Enforcement: Limitations and Solutions

Sarisht Wadhwa, **Luca Zanolini**, Aditya Asgaonkar, Francesco D’Amato, Chengrui Fang, Fan Zhang, Kartik Nayak. ACM CCS 2024. [doi](#)

Recent Latest Message Driven GHOST: Balancing Dynamic Availability With Asynchrony Resilience

Francesco D’Amato, **Luca Zanolini**. IEEE CSF 2024. [doi](#)

Asymmetric Distributed Trust

Orestis Alpos, Christian Cachin, Björn Tackmann, **Luca Zanolini**. *Distributed Computing*, 37(3):247–277, 2024. [doi](#)

A Simple Single Slot Finality Protocol for Ethereum

Francesco D’Amato, **Luca Zanolini**. CBT at ESORICS 2023. [doi](#)

SELECTED TALKS

What Does ePBS Actually Guarantee?

KAIST School of Computing – Invited research seminar

Aug 2026

Daejeon, Korea

The Whys Behind Ethereum’s Consensus Design

BUIDL Asia 2026

Apr 2026

Seoul, Korea

What Comes Next for Institutions on Ethereum

Ethereum Korea One – Closing talk

Apr 2026

Seoul, Korea

Towards a Faster Finality Protocol for Ethereum

PaPoC 2025 – Keynote

Mar 2025

Rotterdam, Netherlands

A Fast Confirmation Rule for the Ethereum Consensus Protocol

Devcon

2024

Bangkok, Thailand

The Ethereum Consensus Protocol

Yale University – Invited talk

2023

New Haven, USA

Recent Latest Message Driven GHOST

a16z Crypto Research Summer School

2023

New York, USA

MENTORING AND COMMUNITY

- Supervised and mentored Ethereum Foundation research interns and an industrial PhD intern; mentored master’s students; provided weekly small-group supervision of University of Cambridge students in Concurrent and Distributed Systems (2025).
- Organized PhD-group research seminars at the University of Bern; co-organized Ethereum Foundation research workshops and cross-institutional protocol discussions.

ACADEMIC SERVICE

Program committees and reviewing: DISC (2022, 2025); OPODIS (2024, 2025); PaPoC (2025); Science of Blockchain Conference (2026); IEEE SRDS (2026); Financial Cryptography (2026); Asiacrypt (2020); *Distributed Computing* (2021).

SKILLS

Distributed algorithms and consensus | Byzantine fault tolerance | Safety, liveness, and reliability | Protocol specification, implementation, and testing | Rust | Go | Python | Solidity | Concurrency | Cryptography | Agentic systems and AI-assisted engineering (Claude Code, OpenAI Codex) | Dev containers | GitHub | LaTeX | Cross-team collaboration across geographies and time zones.